



Practice: Textbook Exercise 1.6.7 (set size via bijections)

In everyday mathematics, the number of elements of a set is understood in an intuitive sense and no definition is usually given. In the logical foundations of mathematics, however, the number of elements is defined via bijections: $|X| = n$ means that there exists a bijection from X to the set $\{1, 2, \dots, n\}$. (Also other, alternative definitions of set size exist but we will consider only this one here.)

- (a) Prove that if X and Y have the same size according to this definition, then there exists a bijection from X to Y .
- (b) Prove that if X has size n according to this definition, and there exists a bijection from X to Y , then Y has size n too.
- *(c) Prove that a set cannot have two different sizes m and n , $m \neq n$, according to this definition. Be careful not to use the intuitive notion of “size” but only the definition via bijections. Proceed by induction.